Physical Oceanography (MO8013), Central Concepts 2026

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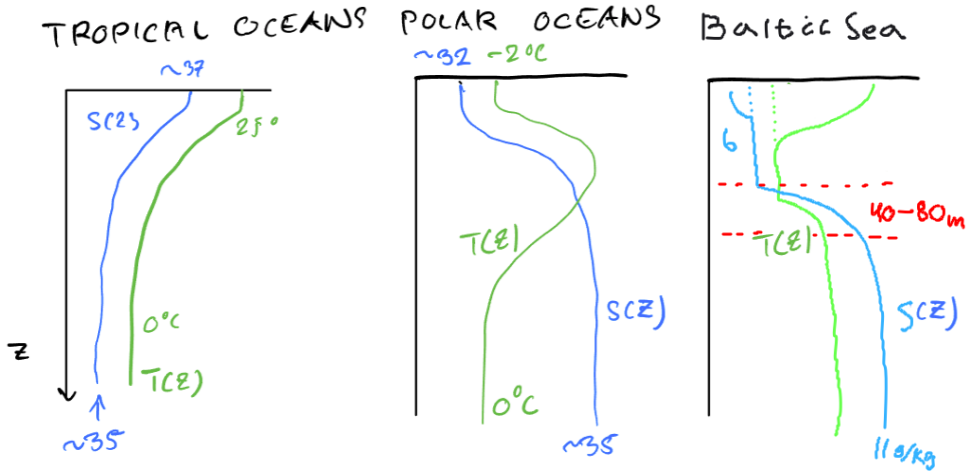
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Physical features of the ocean and relevant dynamical approximations

1. Sea Water: the equation of state $\rho(S,T,P)=\rho_0[1-\beta_T(T-T_0)+\beta_S(S-S_0)+\beta_p(p-p_0)]$. At 25 g/kg, the TMD and freezing point meet. Below we have brakish, above the freezing point is the maximum density.

2. The buoyancy frequency N measures the stability . However the comprassability-term cannot be negelected if one deals with the deep ocean, $N^2 \approx -g\left(\frac{1}{\rho_0}\frac{\partial \rho_0}{\partial z}+\frac{g}{c_s{}^2}\right)\approx -\frac{g}{\rho_0}\frac{\partial \rho_0}{\partial z}$.

3. Sketch vertical T and S profiles in the North Atlantic, the North Pacific, the Arctic Ocean, and the Baltic. Major ocean currents, e.g., the Gulf Stream and Kuroshio.



4. Interpret T –S diagrams: Consider two water masses with different T and S; illustrate in a T–S diagram what happens if the water masses mix. Where can you find the water masses AABW (Antarctic Bottom Water), AAIW (Antarctic Intermediate Water), and NADW (North Atlantic Deep Water)?

5. Write down the hydrostatic dynamical equations on the tangent plane are $\frac{D\vec{u}}{Dt}=\frac{1}{\rho_0}\nabla p-g\hat{z}+\nu\nabla^2\vec{u}$.

6. Hydrostatic applies when the horizontal scales are much larger than vertical scales, Boussinesq applies when the perturbation of density is small.

$$\frac{D\vec{u}_1}{Dt} = -g\nabla_H \eta_1 \approx -g'\nabla_H h_1$$

7. In a two layer model we have $\frac{D\vec{u}_2}{Dt} = -\frac{\rho_1}{\rho_2}\left(g\nabla_H \eta_1 + g'\nabla_H \eta_2\right)$, limit gives $\frac{D\vec{u}_1}{Dt} \approx -g'\nabla_H h_1$

$$\frac{\partial \vec{\eta}_i}{\partial t} + h_i\left(\nabla_H \cdot \vec{u}\right) = 0$$

Ekman

1. The equations governing the Ekman flow are $f\,\hat{z}\times\vec{u} = -\frac{1}{\rho_0}\nabla p + \frac{1}{\rho_0}\frac{\partial \vec{\tau}}{\partial z}$.

2. The horizontal Ekman transport is related to the surface wind stress as $\vec{U}_E = -\frac{1}{\rho_0}\hat{z}\times\left(\frac{\vec{\tau}_0}{f}\right)$.

3. The vertical surface Ekman velocity is related to the surface wind stress as $\vec{W}_E = \frac{1}{\rho_0}\nabla_H \times\left(\frac{\vec{\tau}_0}{f}\right)$.

4. In a *cyclonic* circulation bottom Ekman transport → divergence and upward motion.
In an *anticyclonic* circulation bottom Ekman transport → convergence and downward motion

5. Illustrate costal upwelling and downwelling, and equatorial upwelling using simple sketches. Wind driven gyres (Vallis 14)

Wind driven gyres

1. The Sverdrup relation gives $\beta v H = f w_E$.

2. The Stommel's solution to the wind-driven gyre problem: ad shear stress $\Gamma\frac{\partial v}{\partial x} = \Gamma\frac{\partial^2 \psi}{\partial x^2}$ → $\psi = \psi_{SV}\left[1 - \exp\left(\frac{x}{\delta_s}(x - x_w)\right)\right]$.

Waves: Poincare and Kelvin

1. Given the linearised SWE solution gives Poincare Waves as $\omega^2 = f^2 + c^2k^2$. Assuming $\eta = \eta_0\cos(kx - \omega t)$ and solving for $\frac{v_0}{u_0} = \frac{f}{\omega}$ → internal oscillations when $k \to 0$.

2. Derive Kelvin waves travelling in the positive x- direction along a coast located at y = 0 by assuming $\eta = \eta_0\exp\left(\frac{i}{k}[x \mp \omega t]\right)$ → $\omega = ck$.

3. The features of an equatorial Kelvin Wave are given by $f \approx \beta y$. The speed is either $c_{BT} = \sqrt{gH}$ or $c_{BC} = \sqrt{g' h_1}$.

4. There is a seiche in the Baltic, extending from St Petersburg to southwestern Baltic (around Lübeck, Germany). Suppose that the seiche is a Kelvin wave. The period is given by $T = L/c$.

5. The main lunar tide M_2 has a period of **12.4 hours** and the main solar tide S_2 has period of 12.0 hours.

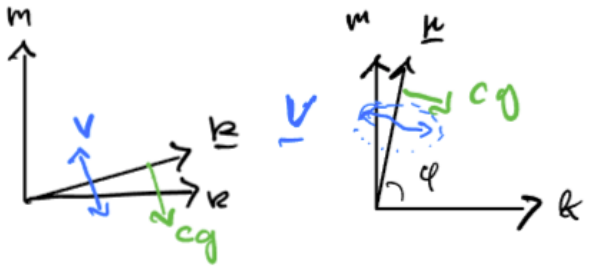
Internal waves and a rotating stratified fluid

These waves are sometimes just referred to internal waves; names including inertial–gravity waves and rotating gravity waves are also used in the literature.

1. The frequency band of internal waves is $N \geq \omega \geq f$, $N \sim 10^{-2}$ s^{-1} and $f \sim 10^{-4}$ s^{-1}. If we have a case where $f > N$ no waves can exsist.

2. The *dispersion relation of internal waves* are detrmined by the aspect ratio (α), which is the ratio between the vertical scale and the horizontal scale. Large α implies $\omega \sim N$; and small α implies $\omega \sim f$.

3. A sketch showing the wavenumber vector, $\vec{k} = (k, m)$, and lines of constant phase for an internal wave with $\omega \sim N$ and $\omega \sim f$. Indicate the phase and group velocities in the diagram.



4. Barotropic tidal currents flowing back and forth over seafloor topo- graphy can generate internal waves, also known as internal tides. The internal tides have the same frequency as the barotropic tide. Use the Poincare wave dispersion relation ω2 = f 2 + c2(k2 + l2) to compute the wave length of the M2 tide (period 12,4 h) for the barotropic tide (use c = 200 m s−1) the internal tide (use c = 2 m s−1), if f = 10−4 s−1. What is the ratio of the wavelengths of the barotropic and the internal tide?

Rossby Waves

Rossby waves are a type of waves that arise from and propagate on a back- ground gradient in potential vorticity; see Vallis sec. 6.3.3 and Eq. (RW.1). Planetary Rossby waves arise from the latitudinal gradient of f , i.e. β = df /dy. Along the continental slopes in the ocean, there are topographic Ross- by waves, propagating along the slope (or coast line), which arise from the depth (H) variations of the background potential vorticity f /H. Vallis section 6.1 and 6.2 summarises some fundamental and general wave features.

1. On the mid-latitude β-plane – f = f0 + βy – the linearised SWE vorticity equation is ∂ζ ∂t + f0∇ · u + βv = 0, (1) (can you show this?) and the continuity equation is ∂η ∂t + H0∇ · u = 0. (2)

Use these equations, and the assumption that flow to the leading order is in geostrophic balance (u, v) = g/f0(−∂η/∂y, ∂η/∂x), and derive the Rossby wave equation.

2. Assume a plane wave, $\eta \propto \exp[i(\omega t - kx - ly)]$, and write down the Rossby wave dispersion relation. [Answer $\omega = -\beta k(L_{-2} D + k2 + l2)_{-1}$.] Sketch the dispersion relation for l = 0.

3. The east–west phase speed of Rossby waves has a specific sign. Which?

4. What is the approximate equation that describe long Rossby waves?

5. Describe qualitatively the Rossby wave propagation mechanism. (Vallis

6. What is the maximum frequency of a Rossby wave? [This occurs where ∂ω/∂k = 0, implying that the east–west group velocity is zero: $\omega max = \beta/(2 \sqrt{} L_{-2} D + l2)$; if l = 0, then $\omega max = LD\beta/2$.]

7. If l = 0 and LD = 30 km, what is the shortest period of Rossby waves at 45 degrees north? What is the answer at 45 degrees south? [Answer: the shortest wave period is given by $4\pi/(LD\beta)$.]

8. Consider a long Rossby wave with westward group velocity. Describe what happens when the wave is reflected at a western ocean boundary. What happens with the wave length and amplitude? Hint: ω is conser- ved in the reflection.

9. Describe qualitatively how the Stommel–Sverdrup gyre develops from an initial state when the wind is switched on. What is the role of the long westward travelling and short eastward travelling Rossby waves? (See Gill section 12 and the lecture notes.)

Baltic Sea

See the lecture notes.

1. An estuary is a semi-enclosed sea, with restricted communication with the ocean and a net supply of freshwater from river run off and precipitation. The Baltic Sea, the Black Sea and the Arctic Ocean are estuaries. Describe the typical circulation in an estuary.

2. Knudsen's relations: Consider a two-layer description of an estuary, with upper (lower) layer salinity and volume flow S1, M1 (S2, M2) and freshwater input F . Write down the relations for conservation of volume and salt. Use this to derive how F/M1 is related to S1 and S2.

3. Sketch a typical salinity profile in the central Baltic. Indicate roughly the depth of the halocline, and give typical surface and bottom salinities.

4. How is the water in the deepest part of the Baltic (Landsort Deep and the Gotland Basin) renewed?

5. The Mediterranean Sea is a basin with a negative freshwater balance (evaporation exceeds precipitation plus river run off). The outflow from the Mediterranean Sea through the Gibraltar

Strait has a two layer character, with inflow in the upper part and outflow in the lower part. Can the Knudsen relation describe the Mediterranean Sea? Is S1 larger or smaller than S2?

Surface gravity waves

See the lecture notes.

1. You stand at beach and observe waves with a period of ten second. What is the wavelength of the waves in the open ocean where the depth is 2500 m? Is the wavelength shorter or longer near the beach? [Hint use the dispersion relation for waves in deep water $\omega = \sqrt{gk}$, and check whether the calculated wavelength is small compared to the depth.]

2. Torö is a surfing spot south of Stockholm, located 500 km from the Polish coast. On a good surfing day at Torö the wave period is around 7 s (on less good days with smaller waves the period may be around 4–5 s; in the Atlantic typical periods are 10 s or longer). How long time does it take for waves of this period to travel from the Polish coast? Is it the phase or the group velocity that decides the travel time? What is the wavelength of the waves? [Hint use the dispersion relation for waves in deep water $\omega = \sqrt{gk}$.]